

Statistical Inference and Modeling

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- Lecture 1: Introduction
- Lecture 2: Model selection
- Lecture 3: Resampling
- Lecture 4: Linear regression
- Lecture 5: Robust regression
- Lecture 6: Classification
- Lecture 7: Logistic regression
- Lecture 8: Neural networks
- Lecture 9: K-nearest neighbor
- Lecture 10: Bayesian decision, LDA, QDA, and naive Bayes
- Lecture 11: Support vector machine
- Lecture 12: Tree-based methods
- Lecture 13: Bagging, random forest, boosting
- Lecture 14: Unsupervised learning and clustering algorithms
- Lecture 15: Unsupervised learning and PCA
- Lecture 16: Gaussian mixture model clustering
- Lecture 17: DBSCAN clustering

ไปหลังที่ใช้งานบ่อยมาก

5. Robust Regression

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- 1 Table of Contents
- 2 Outliers in training data
 - Outliers in measurements (y)
 - Leverage points in input features
- 3 Outlier diagnostics
 - Recap: linear regression
 - Hat matrix (unusual x values)
 - Residual analysis
 - Cook's distance
- 4 Robust regression
 - Weighted least-squares
 - Loss Functions for Robust Regression
 - Regression M-estimators
 - Median absolute deviation
 - Iterative re-weighted least square
- 5 Summary
- 6 Appendix
- 7 Index

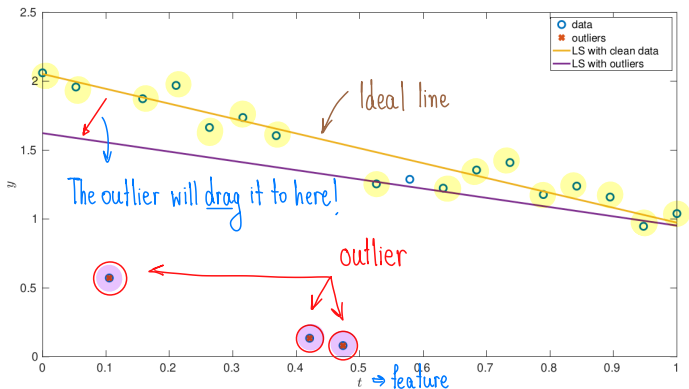
Outliers in training data

- 2 Outliers in training data
 - Outliers in measurements (y)
 - Leverage points in input features

Outliers in training data

↳ The technique which remove noise of measurement. → which extremely diverging.

Problem: The output from linear regression can be impacted by outliers in training data [4, 5].

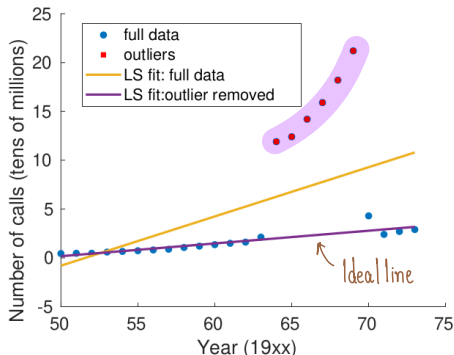


- **Outliers:** data points with extremely diverging measurements. y (measurement)
 - **High leverages:** data points with different mapping between input and output prediction. Usually, the problem lies in the input features. x (input feature)
- Robust regression can overcome the problems.

Outliers in measurements (y)

Example: This is a problem of regression for the number of international phone calls from Belgium during 1950-1975 [4].

- Settings:
 - **Model:** $\#calls = f(year)$
 - **Input** features: year
 - **Output** prediction: the number of calls ($\#calls$)
- Nevertheless, the phone call contains outliers during 1964-1969.
- The slope is affected by the vertical outliers (tilted upward) .

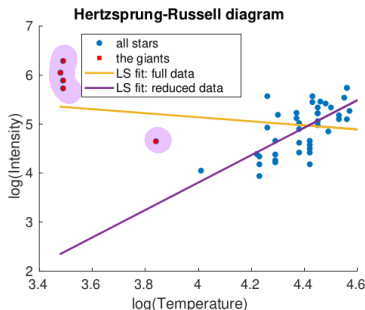


The number of international phone calls from Belgium during 1950-1975 [4]

Leverage points in input features

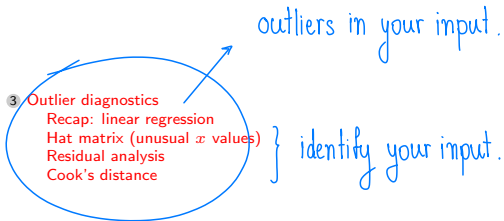
Example: This is a problem of regression for the star intensity [5]

- Settings:
 - **Model:** $\text{intensity} = f(\text{temp})$
 - **Input** features: light intensity (intensity)
 - **Output** prediction: temperature (temp)
- Note that there are 5 giant stars. These stars have significantly lower temperature, which gives results in an inconclusive function mapping.
- Data points such as these are called *bad leverage points*. They are not error, but they contains other input features (such as size of the stars) that can impact the intensity differently from the temperature.



Hertzsprung-Russell diagram of the star cluster CYG OB1 (47 stars) [5]

Outlier diagnostics $\Rightarrow$ มีสมมติฐานว่า y และ x มี linear relationship.



$$\hat{Y} = f(x) = \beta X$$
$$e = E[(\hat{Y} - Y)^2] \text{ (MSE) } \text{จะตัองน้อยๆ.}$$
$$Y = y_0 + \epsilon$$
$$\epsilon \sim N(0, \sigma^2)$$

ในชีวิตจริง เรามีเทคนิคต่าง ๆ มากมาย ในการหา Outlier และวิธีการ
ซึ่งสไลด์นี้เน้นหนึ่งในวิธีที่มี (โบราณแล้ว) และมีมาก feature selection,
cross validation, ... มาช่วยอยู่แล้ว.

How to remove these outliers

Goal: We can use the following methods [4] to identify potentially unusual outliers and remove them... We call these methods **outlier diagnostics**.

Outlier diagnostics: the followings are the methods for checking and removing the outliers in training data:

- Hat matrix
- Residual analysis
- Cook's distance

} result : delete one data point and measure impact.
↳ ลบจุดแล้ว ทำ Regression ใหม่ ต้องดีกว่าเดิม.

Generally, these will compute a regression based on deleting **one data point** and measure **the impact**. The impact is measured as a residual. If the residual is too large, they will be identified as outliers.

Recap: linear regression

Settings: we have m measurements, i.e., $\mathbf{y} \in \mathcal{R}^m$ and $\mathbf{X} \in \mathcal{R}^{m \times n}$ where

$$\mathbf{X} = \begin{bmatrix} (\mathbf{x}^1)^T \\ (\mathbf{x}^2)^T \\ \vdots \\ (\mathbf{x}^m)^T \end{bmatrix} = \begin{bmatrix} x_1^1 & x_2^1 & \dots & x_n^1 \\ x_1^2 & x_2^2 & \dots & x_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ x_1^m & x_2^m & \dots & x_n^m \end{bmatrix}$$

$\mathbf{x}^j \in \mathcal{R}^n$ for n being the number of feature dimension.

Formulation:

$$\mathbf{y} \approx \hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta}$$

where $\boldsymbol{\beta}$ can be estimated by solving the least-square error minimization:

$$\boldsymbol{\beta}_{ls} = \underset{\boldsymbol{\beta}}{\operatorname{argmax}} \quad \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 = \sum_j^m (y_j - \sum_i^n x_i^j \beta_i)^2 \quad (1)$$

- $\mathbf{r} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}$ is called the *residual error*.
- $\boldsymbol{\beta}$ that give the smallest $\|\mathbf{r}\|_2$ is called the *least-squares solution*.
- $\boldsymbol{\beta}_{ls}$ can be obtained from finding $\boldsymbol{\beta}$ that gives the gradient $\nabla_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 = 0$.

$$\nabla_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}_{ls}\|_2^2 = 2\mathbf{X}^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}_{ls}) = 0$$

$$\boldsymbol{\beta}_{ls} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \dots$$

If $\mathbf{X} \in \mathcal{R}^{m \times n}$ with $m \geq n$ is full rank, then the *least-squares solution* $\boldsymbol{\beta}_{ls}$ is unique [2]. Up until this point, both $\mathbf{X}$ and $\mathbf{y}$ are of the training data.

Hat matrix (unusual x values)

Given β_{ls} , one can obtain the prediction $\hat{y}^{\text{Test}}$ by multiplying β_{ls} with the input feature X^{Test} . Note that we use superscript 'Test' to separate the training data components that is used to derive β_{ls} .

$$\hat{y}^{\text{Test}} = X^{\text{Test}} \beta_{ls} = X^{\text{Test}} \beta_{ls} = X^{\text{Test}} \underbrace{(X^T X)^{-1} X^T}_{\text{Train}} y$$

Estimated β

have noise form matrix y .

The hat matrix of a linear regression is defined as

$$H = X^{\text{Test}} \underbrace{(X^T X)^{-1} X^T}_{\text{Train}}$$

that maps the measurements y (from training data).

The matrix form can be written as follows:

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_{m'} \end{bmatrix}^{\text{Test}} = \underbrace{X^{\text{Test}}}_{m' \times n} \underbrace{(X^T X)^{-1} X^T}_{\text{Train}} y = \begin{bmatrix} h_{1,1} & h_{1,2} & \dots & h_{1,m} \\ h_{2,1} & h_{2,2} & \dots & h_{2,m} \\ \vdots & \vdots & \dots & \vdots \\ h_{m',1} & h_{m',2} & \dots & h_{m',m} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$$

$m' \times n$ $(n \times m)(m \times n)$ $(n \times m)$

$m' \times m$ $m \times 1$

measurements from train set.

- $h_{j,i}$ transfers the impact from the i th observation to $\hat{y}_j$.
- If we take derivative $\partial \hat{y}_j / \partial y_i$, then $h_{j,i}$ gives the effect of y_i to $\hat{y}_j$.
- Because H is a matrix that is divided by $X^T X \in \mathbb{R}^{n \times n}$, it is possible that the diagonal coefficient $0 \leq h_{j,j} \leq 1$.

Hat matrix (unusual x values)

ใช้ในการตรวจสอบ input feature.

H-matrix

$$H = X^{\text{test}} (X^{\text{train}})^{-1} X^{\text{train}T}$$

ideal case $H \approx I$.

ถ้าเรามองว่า X^{test} เป็น X^{train} $\Rightarrow H = X(X^T X)^{-1} X^T \approx I$ Ideal case.

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_{m'} \end{bmatrix}^{\text{Test}} = X^{\text{Test}} \underbrace{(X^{\text{Train}T} X^{\text{Train}})^{-1} X^{\text{Train}T}}_{\text{Train}} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$$

Diagram illustrating the matrix multiplication with annotations: A blue box highlights the diagonal elements $h_{1,1}, h_{2,2}, \dots, h_{m',m'}$ in the hat matrix, which are close to 1. Red lines and circles highlight the off-diagonal elements, which are close to 0.

กรณีที่ $\hat{y}^{\text{test}}$ มีค่าห่างจาก y^{test} มากๆ แปลว่า β มีปัญหา

$X^{\text{train}} = X$ ไม่ดีพอในการ Train.

หรือ y^{train} ไม่ค่อย match กับ y^{test} $\Rightarrow$ out-of-distribution.

$\therefore$ ถ้าเกิด H-matrix $\neq I$ หรือ y^{train} และ y^{test} ต่างกันมาก $\Rightarrow$ เกิด High leverage point.

ไม่ได้หมายถึง y ผิด
แต่ X ของจุดนั้นแปลกเมื่อเทียบกับ train

Example

One can calculate the averaged value of diagonal coefficients in matrix H as follows:

ค่าเฉลี่ยของ diagonal line $\rightarrow \bar{h} = \sum_j h_{j,j} / \min(\{m', m\})$

diagonal element.

จำนวน training set.

จำนวน testing set

which should be bounded by n/m' , i.e., $\bar{h} \leq n/m'$.

First, $\min(\{m', m\}) = m'$ because it is common that the numbers of observation at testing phase will be less than the one's at training.

But what about the denominator n ? Let's see $X \in \mathcal{R}^{m \times n}$. One can show the followings [3]:

$$\sum_j h_{j,j} = \text{Tr}(H) = \text{Tr}(\underbrace{X^{\text{Test}} (X^T X)^{-1} X^T}_{\text{Train}}) \approx \text{Tr}(X^T X (X^T X)^{-1}) = \text{Tr}(I_{n \times n}) = n$$

Trace

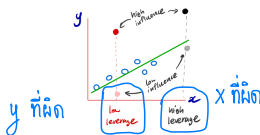
diagonal เาคาดหวังว่า $H \approx I_{n \times n}$

Then, we define **high leverage data points** as any $h_{j,j}$ that are larger than the averaged value by some threshold:

The j^{th} data pair (x^j, y_j) is an **high leverage**, if $h_{j,j} > \frac{2n}{m'}$.

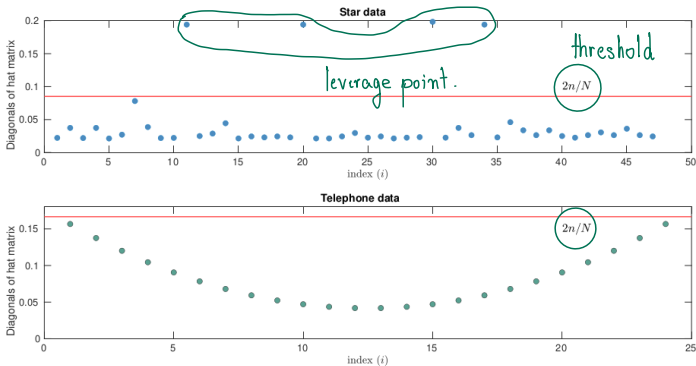
Nevertheless, the high leverage data points can have either **low** or **high** influence on the error:

- they have **low** influence on the error, if y_j deviates from the norm by a small distance.
- they have **high** influence on the error, if y_j greatly deviates from the norm.



Example

The diagonal values of hat matrix (star and telephone datasets)
 X is standardized. Red lines are the thresholding value at $2n/m'...$



- $h_{j,j}$'s of the giant stars are larger than $2n/m'$ ($2n/m'=0.085$).
- $h_{j,j}$'s of the telephone data are all under $2n/m'$, so no outliers are pointed out.
- Hat matrix is a function of X can detect the outliers in **input features**.
- Whether or not the data point is influential depends on y_i .

Idea of residual analysis

$$\hat{y}^{\text{Test}} = X^{\text{Test}} \beta_{ls} - H y \rightarrow \hat{y}^{\text{Test}} \approx y \leftrightarrow H \approx I$$

Let us consider H in the outlier analysis. In the ideal case, X^{Test} will be similar to the training's, that is, $X^{\text{Test}} \approx X$ in the hope that $H \approx I$

ไม่แน่ใจโตจุดหนึ่งมี leverage สูงกว่าปกติ.
(เมื่อเทียบกับจุดอื่น.)

$$H = X^{\text{Test}} (X^T X)^{-1} X^T \approx I \quad (2)$$

As a side note, one should be able to calculate a leverage h_{jj} by $h_{j,j} = \underbrace{(x^j)^T}_{\text{Test}} \underbrace{(X^T X)^{-1} (x^j)}_{\text{Train}}$.

- Eq.(2) appears similar to psuedo matrix inversion, which should guarantee that $H = I$. However, this may not be the case, especially if the input features are not a good representative for its predictive value $\hat{y}$, or those testing input features are different from the training input features.
- Therefore, another way to diagnose for the outliers is to consider the residual from the identity matrix:

$$e = y - \hat{y} = (I - H)y \quad \text{ทุกอย่างอยู่บน training.} \quad (3)$$

- Nevertheless, this approach lacks the consideration on the scale of y . In the followings, we will explore two more concrete approaches to remove the outliers based on the residual vector d .

Residual analysis ใช้ β จาก training แล้วหา Residual ใน train/valid ได้

An alternative methods for checking whether the j^{th} data pair (x^j, y_j) is an outlier:

- Compute $\beta_{ls}^{-j} \in \mathcal{R}^{N \times M-1}$ with the remaining data, i.e.,

$$\beta_{ls}^{-j} = (\overset{n \times n}{X^T X})^{-1} \overset{n \times (M-1)}{X^{-jT}} \overset{(M-1) \times n}{y^{-j}}$$

- Compute the fitted value after removing y_j by $\hat{y}^{-j} = X\beta_{ls}^{-j}$ where $X \in \mathcal{R}^{m \times n}$ which is the same as the training data.
- Let's define $\hat{y} = X\beta_{ls}$. One can compute the residual between the one with fully mapped data and the one with deleting pairs as follows:

$$d^{-j} = X\beta_{ls} - X\beta_{ls}^{-j} = \hat{y} - \hat{y}^{-j}.$$

- For each coefficient d_i^{-j} (where $d^{-j} = [d_i^{-j}]_{i=1:m}$ and $d_i^{-j} := \hat{y}_i - \hat{y}_i^{-j}$), where $\hat{y}^{-j} = [\hat{y}_i^{-j}]_{i=1:m}$, and $\hat{y} = [\hat{y}_i]_{i=1:m}$. The standardized residual is defined as:

$$r_i^{-j} = \frac{d_i^{-j}}{\text{RSE}(d_i^{-j})}.$$

where $\text{RSE}(d_i^{-j})$ is the residual standard error of d_i^{-j} : $\text{RSE}(d_i^{-j}) = \sqrt{\frac{\sum_{i=1}^m (\hat{y}_i - \hat{y}_i^{-j})^2}{m-n}}$. Here, we use $m-n$ for n is the number of parameters in the linear regression.

- Potential outliers are the one with $r_i^{-j} > 3$ (threshold).

Residual analysis

$$\textcircled{1}. \beta_{ls}^{-j} \Rightarrow \beta_{ls} = (X^T X)^{-1} X^T y$$
$$\beta_{ls}^{-j} = (X^T X^{-1}) X^{-jT} y^{-j}$$

$$X = \begin{bmatrix} \text{remove row} \\ \vdots \end{bmatrix}_{n \times n} \xrightarrow{\text{remove row}} X^{-j} = \begin{bmatrix} \vdots \end{bmatrix}_{(n-1) \times n}$$

$$y = \begin{bmatrix} A \\ c \end{bmatrix} \rightarrow y^{-j} = \begin{bmatrix} A \\ c \end{bmatrix}$$

$$\textcircled{2}. \hat{y}^{-j} = X \beta_{ls}^{-j}; X \text{ ภาวะมาจาก train / valid ก็ได้} \Rightarrow y.$$

$$\textcircled{3}. r^{-j} = \frac{\hat{y} - \hat{y}^{-j}}{\text{Normalize}} \rightarrow \text{ถ้า } \hat{y} - \hat{y}^{-j} \text{ มีค่ามาก ๆ แสดงว่า } (x^j, y^j) \text{ สัมควรเอาออก}$$

$\hat{y} = X \cdot \hat{\beta}$

$\hat{y} = \begin{bmatrix} \text{ } \end{bmatrix}_{n \times 1} = \begin{bmatrix} \text{ } \end{bmatrix}_{n \times n} \begin{bmatrix} \text{ } \end{bmatrix}_{n \times 1}$

$\hat{y}^{-j} = \begin{bmatrix} \text{ } \end{bmatrix}_{n \times 1} = \begin{bmatrix} \text{ } \end{bmatrix}_{n \times n} \begin{bmatrix} \text{ } \end{bmatrix}_{n \times 1}$

j มีอิทธิพลสูง (high influence)

ต่างกันตรงที่ β_{ls} คือเราเทรนปกติ

แต่ β_{ls}^{-j} คือ "—" หลังตัด sample j ออกไป.

Outlier diagnostics — Cook's distance $\Rightarrow$ จุดนี้ทำให้โมเดลเปลี่ยนไปหรือไม่?

Settings... H -matrix $\Rightarrow$ ถ้าเปลี่ยนตำแหน่งจุดหนึ่งใน X -space จะส่งผลต่อเส้น Regression แค่ไหน?

- The estimated parameters after deleting (x^j, y_j) are $\beta_{ls}^{-j} \in \mathcal{R}^{n \times 1}$
- The fully mapped target $\hat{y} = X^{\text{Test}} \beta_{ls} = X \beta_{ls}$
- The mapped target after deleting (x^j, y_j) : $\hat{y}^{-j} = X^{\text{Test}} \beta_{ls}^{-j} = X \beta_{ls}^{-j}$
- The residual from the fully mapped target $d^{-j} = \hat{y} - \hat{y}^{-j} \in \mathcal{R}^{m \times 1}$

Cook's distance measuring how far β_{ls} changes when (X_i, y_i) is removed:

$$D_j = \frac{\sum_{i=1}^m (\hat{y}_i - \hat{y}_i^{-j})^2}{n \cdot \left(\frac{\sum_{i=1}^m (\hat{y}_i - \hat{y}_i)^2}{m-n} \right)}$$

$$\begin{aligned} D_j &= \frac{(\hat{y} - \hat{y}^{-j})^T (\hat{y} - \hat{y}^{-j})}{n \cdot \text{MSE}_y(\hat{y})} \\ &= \frac{(X \beta_{ls} - X \beta_{ls}^{-j})^T (X \beta_{ls} - X \beta_{ls}^{-j})}{n \cdot \text{MSE}_y(\hat{y})} \\ &= \frac{(\beta_{ls} - \beta_{ls}^{-j})^T (X^T X) (\beta_{ls} - \beta_{ls}^{-j})}{n \cdot \text{MSE}_y(\hat{y})} \end{aligned}$$

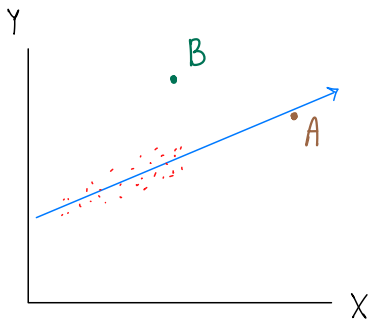
$$\textcircled{1} \hat{y}^{-j} \approx \hat{y} \rightarrow D_j \approx 0$$

$$\textcircled{2} \hat{y}^{-j} \approx y \rightarrow D_j \approx \frac{m-n}{n} \approx 1$$

$m \gg n$

- Note that the denominator is changed to $n \cdot \text{MSE}_y(\hat{y})$ where n is the rank of X .
- $\text{MSE}_y(\hat{y})$ is the (empirical) mean square error of the regression model: $\text{MSE}_y(\hat{y}) = \frac{e^T e}{m-n}$, where e is the residual vector defined in 3.
- D_j measures the effect of the j th observation on the fitted vector and coefficients. High D_j indicates that the observation could be outlier (threshold can be $4/n$ or $3 \cdot \bar{D}$)

Outlier diagnostics — Cook's distance



จุด A $\Rightarrow$ ห่างจาก X อื่นๆ มาก แต่
เส้น Regression ผ่านใกล้มากๆ (e ต่ำ)

$\therefore$ H-matrix สูง, Cook ต่ำ

$\Rightarrow$ จุดนี้อาจอันตราย แต่ยังไม่เกิดปัญหา

จุด B $\Rightarrow$ ไม่ได้ใกล้มาก แต่ y ผิดพลาดสูง

$\therefore$ H-matrix ปานกลาง, Cook สูง.

$\Rightarrow$ จุดนี้จะสร้างปัญหา แก่โมเดลแน่ๆ.

Cook's distance

The cook's distance

$$D_j = \frac{(\beta_{ls} - \beta_{ls}^{-j})^T (\mathbf{X}^T \mathbf{X}) (\beta_{ls} - \beta_{ls}^{-j})}{n \cdot \text{MSE}_y(\hat{y})} \quad (4)$$

has an equivalent form, which is expressed using the leverage ($0 \leq h_{j,j} \leq 1$)

$$D_j = \frac{e_j^2}{n \cdot \text{MSE}_y(\hat{y})} \frac{h_{j,j}}{(1 - h_{j,j})^2} \quad \text{ข้อมูลทั้งหมด มาจาก training.} \quad (5)$$

where e_j is a coefficient in the residual vector $\mathbf{e} = [e_j]_{j=1:m} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y}$. The proof that Eq.(4) is equivalent to Eq.(5) is provided in Appendix (P. 33).

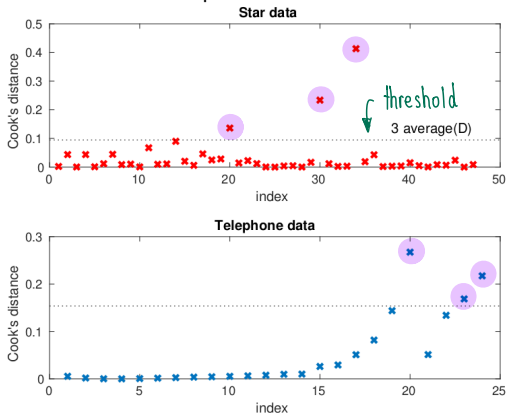
Why you need to know this ...?

#features

- Eq.(5) is more efficient to compute. You can verify this by flop counts.
- Because the impact of the bad leverages will reflect through $\hat{\mathbf{y}}$, but the measurement outliers will further extend the difference between $\mathbf{y}$ and $\hat{\mathbf{y}}$. Note that both will contribute to the high D_j as shown in Eq.(4).
- However, Eq.(5) separates the impact of measurement outliers from the impact of the bad leverages (input feature outliers). This gives a better view towards identifying problems in the linear regression.

Example of Cook's distance

Cook's distance calculated on star and telephone datasets:

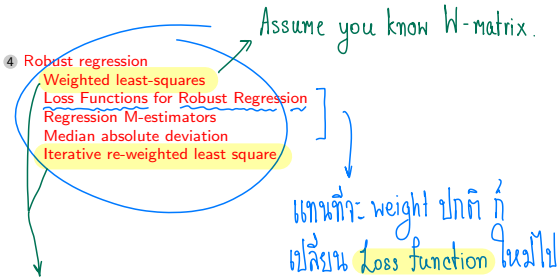


- star indices of high D_i are 20, 30, 34 (three of the giant stars).
- telephone indices of high D_i are 20, 23, 24 (years 1969, 1972, 1973).

Robust Regression / Robust Estimation. $\rightarrow$ tool ที่ทำให้ Model เราเพิ่มความ Robustness

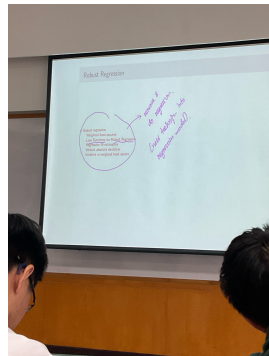
$\rightarrow$ ลด effect จาก training sample ที่มี noise ปนอยู่.
(เราพยายามให้ data ทุก sample ไม่อยากติดอะไรออกเลย)

ต่อ Noise โดยไปปรับที่ Loss function.



มี Loss function หลักเป็น Least Square.

$\rightarrow$ เอา outlier ออกแล้ว Estimate ใหม่ต่อไป.



Weighted least-squares ***

↳ Weighted matrix

ให้ weight กับ $X\beta - y$ ตามความสำคัญ
Formula ไม่ต่างจากเดิมมาก

Settings: we have m measurements, i.e., $y \in \mathcal{R}^m$ and $X \in \mathcal{R}^{m \times n}$ where

$$X = \begin{bmatrix} (x^1)^T \\ (x^2)^T \\ \vdots \\ (x^m)^T \end{bmatrix} = \begin{bmatrix} x_1^1 & x_2^1 & \dots & x_n^1 \\ x_1^2 & x_2^2 & \dots & x_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ x_1^m & x_2^m & \dots & x_n^m \end{bmatrix}$$

$x^j \in \mathcal{R}^n$ for n being the number of feature dimension.

The weighted least-squares problem is defined as follows:

Let W be a positive definite matrix that can be factorized as $W = L^T L$. A weighted least-square (WLS) problem is

$$\beta_{ls} = \underset{\beta}{\operatorname{argmin}} (X\beta - y)^T W (X\beta - y) = \|L(X\beta - y)\|^2$$

เราต้องกำหนด W เอง $\rightarrow$ ปัญหา!

You can use L to suppose the unwanted pairs.

Similar to the simple linear regression, we can solve for β by applying gradient with respect to β , and equate the equation to zero.

$$2X^T L^T L (X\beta - y) = 0$$

$$X^T W X \beta = X^T W y$$

$$\beta_{wls} = (X^T W X)^{-1} X^T W y \quad (\text{if } X \text{ is full rank}).$$

* Assume that you know *
which pairs could be bad.

→ Rely on this assumption

Interpretation of weighted least-squares

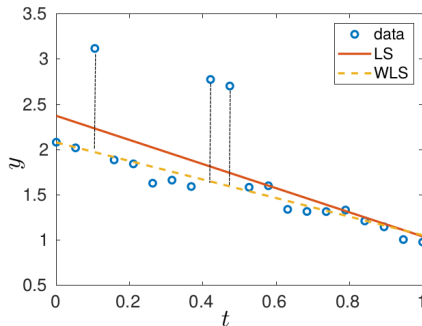
เราต้องกำหนด w เอง.

As samples 3, 9, 10 are outliers, we can suppress them by applying

$\mathbf{W} = \text{diag}(w_1, w_2, \dots, w_m) = \text{diag}(l_1^2, l_2^2, \dots, l_m^2)$ onto the weighted least-square objective:

$$\sum_{j=1}^m w_j (y_j - \mathbf{x}^j \beta)^2.$$

- Use relatively low w_3, w_9, w_{10} to penalize samples 3, 9, 10.
- The weights can adapt to the outliers which improves the robustness over the linear regression (denoted as the ordinary least square or LS).



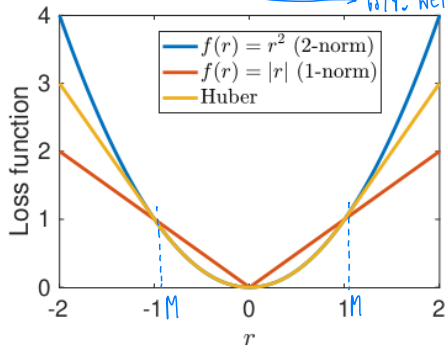
Loss functions for Robust Regression

$$\underline{r} = y - X\beta$$

- l_2 -norm has the highest penalty on the large residual $|r| > 1$.
- However, the residual, if too large, will dominate the objective loss, which leads to the incorrect regression.
- Instead Huber and l_1 norms are often used in robust regression.
- These losses (l_2 , l_1 , Huber) can all be formulated as convex problem.

→ การทำ least square มันกำลังลง r หหมดเลย ไม่เห็นผลมากนัก เพราะจะเกิด effect ในที่, งามมองอื่นๆ ไม่เห็น.

→ เพราะ weight r เท่ากันหมด



$$r = y - X\beta$$

ขั้นบันได.

$$\text{Huber}(r) = \begin{cases} (1/2)r^2, & |r| \leq M \\ M(|r| - M/2), & |r| > M \end{cases}$$

Loss f^n : $\underset{\beta}{\operatorname{argmin}} \|y - X\beta\|_{l_1}$

$\underset{\beta}{\operatorname{argmin}} \text{Huber}(r)$

Regression M-estimators $\Rightarrow$ เปลี่ยน loss ฟังก์ชันของ deviation

Generally, the regression model can be generalized into the M-estimator regression formulation defined as follows

$$\beta_M = \underset{\beta}{\operatorname{argmin}} \sum_{j=1}^m \rho \left(\frac{r_j(\beta)}{\hat{\sigma}} \right) \quad (6)$$

$r_j(\beta) = y_j - x_j^T \beta$
Absolute deviation (AD) $\rightarrow \hat{\sigma}$ for all j

where $\hat{\sigma}$ is a predefined scale.

- For $\rho(r) = r^2$, the M-estimator Eq.(6) becomes a least-square problem, and solving this equation will give $\beta_M = \beta_{ls}$.
- For $\rho(r) = |r|$, the M-estimator Eq.(6) becomes the least absolute deviation (LAD)

$$\beta_{lad} = \underset{\beta}{\operatorname{argmin}} \sum_{j=1}^m \frac{|y_j - x_j^T \beta|}{\hat{\sigma}}$$

where solving the above equation will give $\beta_M = \beta_{lad}$. Here, $\hat{\sigma}$ is the predefined scale value.

- The predefined scale can be estimated using median absolute deviation:

$$\hat{\sigma} = k \cdot \operatorname{MAD}_{y_j}(\hat{y}_j)$$

where k is the scale factor ≈ 1.4826 , and MAD is shorten for the median absolute deviation function $\operatorname{MAD}_{y_j}(\hat{y}_j) := \operatorname{Median}(|\hat{y}_j - y_j|)$ where $\hat{y}_j = x_j^T \beta_M$ and y_j is the reference.

Median absolute deviation

Let $\hat{x}^j$ be an estimate to the data point x^j in $\{x^1, x^2, \dots, x^m\}$. The median absolute deviation (MAD) function from its median is defined as:

$$\text{MAD}_{\tilde{x}}(x^j) = \text{Median}(|x^j - \tilde{x}|)$$

where $\tilde{x}$ is the median of the set $\{x^1, x^2, \dots, x^m\}$.

To use MAD as the estimator, one have to multiply a scale factor k to the median absolute deviation (MAD) function:

$$\hat{\sigma} = k \cdot \text{MAD}_{\tilde{x}}(x^j)$$

Some properties:

- Median absolute deviation (MAD) function is a robust estimation function for the scale parameter.
- k is the scale factor, which depends on the data distribution. For a normally distributed data, k is taken to be $k = \frac{1}{\Phi^{-1}(3/4)} \approx 1.4826$.
- Derivatoin: (next) how do we know that $k = \frac{1}{\Phi^{-1}(3/4)} \approx 1.4826$. ?

Median absolute deviation (scale factor k)

To use MAD as the estimator for standard deviation ($\hat{\sigma}$), one have to multiply a scale factor k to the median absolute deviation (MAD) function:

$$\hat{\sigma} = k \cdot \text{MAD}$$

Assume $\hat{\sigma}$ is $N(0,1)$

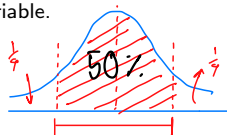
For any normally distributed data, $k = \frac{1}{\Phi^{-1}(3/4)} \approx 1.4826$.

Below ... we show why

- Scale factor k is $1/\text{Boundary}$, where the Boundary is the deviated value that cover $x\%$ of the data distribution.
- Let's define the boundary $\pm \text{MAD}$ that covers 50% of the standard normal distribution (between $1/4$ and $3/4$). Suppose that X is any random variable.
- The normalized variable $Z = \frac{X - \mu}{\sigma}$

$$P(Z \leq \frac{\text{MAD}}{\sigma}) = P\left(\frac{|X' - \mu|}{\sigma} \leq \frac{\text{MAD}}{\sigma}\right)$$

$$= \Phi\left(\frac{\text{MAD}}{\sigma}\right) - \Phi\left(-\frac{\text{MAD}}{\sigma}\right) = 2\Phi\left(\frac{\text{MAD}}{\sigma}\right) - 1 = \frac{1}{2}$$



We use the fact that $\Phi(-x) = 1 - \Phi(x)$.

- Then, $\Phi\left(\frac{\text{MAD}}{\sigma}\right) = 3/4$, and $\frac{\text{MAD}}{\sigma} = \Phi^{-1}(3/4) = 0.67449$.
The scale factor k is

$$k = \frac{1}{\Phi^{-1}(3/4)} \approx 1.4826$$

Iterative re-weighted least square

ปรับ β ไปเรื่อย ๆ โดยใช้ Weight Matrix, H matrix วนคิด
with cook distance.
 $W = \text{diag}(w_j)$

Iterative re-weighted least square: an iterative method that is used for solving the weighted least-square [1]:

$$\beta^{(t+1)} = \underset{\beta}{\text{argmin}} \sum_{j=1}^m w_j^{(t)} (x_j^T \beta^{(t)} - y_j)^2$$

where $\beta^{(t+1)}$ is updated in each iteration. The weight $w_j^{(t)}$ is also updated in a way of mitigating the influence of outliers.

$$W = \begin{bmatrix} w_1 & 0 & & 0 \\ 0 & w_2 & & 0 \\ & & \ddots & \\ 0 & & & w_m \end{bmatrix}$$

$$\beta^{(t+1)} = \underset{\beta}{\operatorname{argmin}} \sum_{j=1}^m w_j^{(t)} (\mathbf{x}^j{}^T \beta^{(t)} - y_j)^2$$

Generally, the iterative process consists of four steps:

- ① Estimate β^t with a linear regression and compute the residual.
- ② Find the high leverage coefficients and estimate the standard deviation of the residual.
- ③ Compute the weights to suppress the effects of these coefficients.
- ④ Recompute $\beta^{(t+1)}$ with the new weights.

Iterative re-weighted least square process (details) [5]

- 1 Start with calculating $\beta^{(t)}$ and the residual vector

$$\beta^{(t)} = (X^T X)^{-1} X^T y$$

$$\mathbf{r} = [r_j]_{j=1:m}$$

where $r_j = \mathbf{x}_j^T \beta^{(t)} - y_j$. for all j in training sample

- 2 Compute the leverage values $\mathbf{h} = [h_{j,j}]_{j=1:m} = \text{Diag}(\mathbf{H})$ where $\mathbf{H} = X\mathbf{X}^T$.
- 3 Compute a normalized residual. There are two options to calculate the residual:

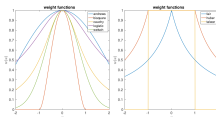
- $u_j = \frac{r_j}{\sqrt{1-h_{j,j}}}$
- $u_j = \frac{r_j}{K \hat{\sigma} \sqrt{1-h_{j,j}}}$

where K is a tuning constant, and $\hat{\sigma} := k \cdot \text{MAD}_{y_j}(\hat{y}_j)$ and $k = 1.4826$.

- 4 Compute the robust weights w_i as a function of u_j (larger u_j , smaller w_j).
Bisquare is an example of the robust weight function. It puts zero weight on large residual.

Bisquare: $w_j = \begin{cases} (1 - u_j^2)^2, & |u_j| < 1 \\ 0, & |u_j| \geq 1 \end{cases}$

$$\mathbf{W} = \text{diag}(w_1, \dots, w_m)$$



Iterative re-weighted least square

5 Update $\beta^{(t+1)}$ by solving weighted least square estimation.

$$\underset{\beta}{\operatorname{argmin}} \sum_{j=1}^m w_j^{(t)} (\mathbf{x}^j{}^T \beta - y_j)^2$$

Then, we can update $\beta^{(t+1)}$ with $\beta^{(t+1)} = (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{y}$

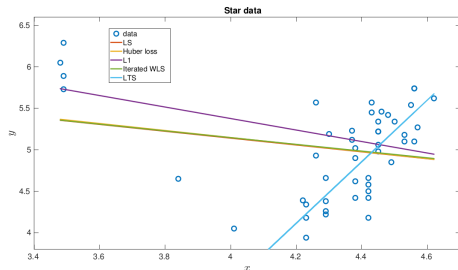
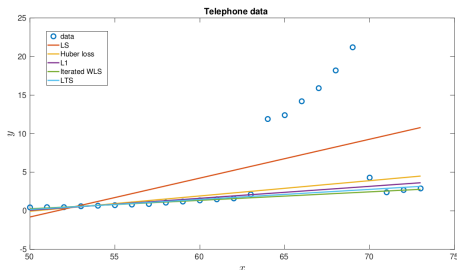
6 Update $\hat{y}_j$ with new $\beta^{(t+1)}$, i.e., $\hat{y}_j = \mathbf{x}^j{}^T \beta^{(t+1)}$ for $j = 1, \dots, m$.

7 Calculate the error.

$$e = \sum_{j=1}^m (\hat{y}_j - y_j)^2.$$

8 Iteration stops if the fit converges, or the maximum number of the iterations is reached, go to the second step.

Result using robust methods



- Telephone dataset has outliers in y -direction; robust methods can ameliorate the effects of outliers quite well
- Star dataset has bad leverage points; most robust methods (WLS, huber, L1) do not perform well on this dataset.
- LTS (least trimmed square estimator) finds a subset of observations with small residuals and estimate $\hat{\beta}$ based on that subset of data (more detail in P. J. Rousseeuw book)

- We consider two important factors that can lower the performance of linear regression:
 - ① **measurement outliers**
 - ② **bad leverage** (input feature outliers)
- There are three main approaches to handle such outliers:
 - ① **outlier analysis** (outlier removal methods): hat matrix, residual analysis, Cook's distance
 - ② **robust estimation methods**: weighted least square, M-estimator.
 - ③ **iterative method**: iterative re-weighted least square
- **Outlier analysis (outlier removal methods)** checks for outliers and remove them.
- **Robust estimation methods** uses loss functions that put less emphasize on the outliers.
- **Iterative re-weighted least square** adjusts the weight function according to fitted residual in each step
- Other methods/algorithms exist: least median square (LMS), least trimmed squares estimator (LTS), random sample consensus (RANSAC)
- Python coding:
<https://github.com/GabbySuwichaya/Statistical-Learning-EE575/blob/master/Tutorial2/main.ipynb>

Appendix: expressing cook's distance using the leverage

The proof that Eq.(4) is equivalent to Eq.(5) is provided in Appendix, i.e.,

$$D_j = \frac{(\beta_{ls} - \beta_{ls}^{-j})^T (\mathbf{X}^T \mathbf{X}) (\beta_{ls} - \beta_{ls}^{-j})}{n \cdot \text{MSE}_y(\hat{y})} = \frac{e_j^2}{n \cdot \text{MSE}_y(\hat{y})} \frac{h_{j,j}}{(1 - h_{j,j})^2}$$

where e_j is a coefficient in the residual vector $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y}$.

We can partition $\mathbf{X}^T \mathbf{X} \in \mathcal{R}^{n \times n}$ as follows:

$$\mathbf{X}^T \mathbf{X} = \sum_{i=1}^m \mathbf{x}^i \mathbf{x}^{iT} = \mathbf{X}^{-jT} \mathbf{X}^{-j} + \mathbf{x}^j \mathbf{x}^{jT}$$

Similarly, we can partition $\mathbf{X}^T \mathbf{y} \in \mathcal{R}^{n \times n}$ as follows:

$$\mathbf{X}^T \mathbf{y} = \sum_{i=1}^m \mathbf{x}^i y_i = \mathbf{X}^{-jT} \mathbf{y}^{-j} + \mathbf{x}^j y_j$$

$$\text{Also ... } \mathbf{X} = \begin{bmatrix} (\mathbf{x}^1)^T \\ (\mathbf{x}^2)^T \\ \vdots \\ (\mathbf{x}^m)^T \end{bmatrix} = \begin{bmatrix} x_1^1 & x_2^1 & \dots & x_n^1 \\ x_1^2 & x_2^2 & \dots & x_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ x_1^m & x_2^m & \dots & x_n^m \end{bmatrix}, \mathbf{X}^{-j} = \begin{bmatrix} (\mathbf{x}^1)^T \\ (\mathbf{x}^2)^T \\ \vdots \\ (\mathbf{x}^{m-1})^T \end{bmatrix}, \mathbf{y}^{-j} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{m-1} \end{bmatrix}$$

Expressing cook's distance ... (cont')

Let's consider $(\beta_{ls} - \beta_{ls}^{-j})$

$$\begin{aligned}\beta_{ls} - \beta_{ls}^{-j} &= \beta_{ls} - \left(\mathbf{X}^{-jT} \mathbf{X}^{-j}\right)^{-1} \mathbf{X}^{-jT} \mathbf{y}^{-j} \\ &= \beta_{ls} - \left(\mathbf{X}^{-jT} \mathbf{X}^{-j}\right)^{-1} \left(\mathbf{X}^T \mathbf{y} - \mathbf{x}^j y^j\right) \\ &= \beta_{ls} - \left(\mathbf{X}^{-jT} \mathbf{X}^{-j}\right)^{-1} \mathbf{X}^T \mathbf{y} + \left(\mathbf{X}^{-jT} \mathbf{X}^{-j}\right)^{-1} \mathbf{x}^j y^j\end{aligned}$$

Compute $\left(\mathbf{X}^{-jT} \mathbf{X}^{-j}\right)^{-1}$ using Sherman–Morrison formula:

$$\begin{aligned}\left(\mathbf{X}^{-jT} \mathbf{X}^{-j}\right)^{-1} &= \left(\mathbf{X}^T \mathbf{X} - \mathbf{x}^j \mathbf{x}^{jT}\right)^{-1} \\ &= \left(\mathbf{X}^T \mathbf{X}\right)^{-1} + \frac{\left(\mathbf{X}^T \mathbf{X}\right)^{-1} \mathbf{x}^j \mathbf{x}^{jT} \left(\mathbf{X}^T \mathbf{X}\right)^{-1}}{1 - \mathbf{x}^{jT} \left(\mathbf{X}^T \mathbf{X}\right)^{-1} \mathbf{x}^j} \\ \left(\mathbf{X}^{-jT} \mathbf{X}^{-j}\right)^{-1} \mathbf{x}^j &= \left(\mathbf{X}^T \mathbf{X}\right)^{-1} \mathbf{x}^j + \frac{\left(\mathbf{X}^T \mathbf{X}\right)^{-1} \mathbf{x}^j \mathbf{x}^{jT} \left(\mathbf{X}^T \mathbf{X}\right)^{-1} \mathbf{x}^j}{1 - \mathbf{x}^{jT} \left(\mathbf{X}^T \mathbf{X}\right)^{-1} \mathbf{x}^j} \\ &= \left(\mathbf{X}^T \mathbf{X}\right)^{-1} \mathbf{x}^j + \frac{\left(\mathbf{X}^T \mathbf{X}\right)^{-1} \mathbf{x}^j h_{j,j}}{1 - h_{j,j}} \\ &= \frac{\left(\mathbf{X}^T \mathbf{X}\right)^{-1} \mathbf{x}^j}{1 - h_{j,j}}\end{aligned}$$

Let's go back ...

$$\beta_{ls} - \beta_{ls}^{-j} = \beta_{ls} - \left(\mathbf{X}^{-jT} \mathbf{X}^{-j} \right)^{-1} \mathbf{X}^T \mathbf{y} + \left(\mathbf{X}^{-jT} \mathbf{X}^{-j} \right)^{-1} \mathbf{x}^j y_j \quad (7)$$

Thus, we have $\left(\mathbf{X}^{-jT} \mathbf{X}^{-j} \right)^{-1} \mathbf{x}^j y_j = \frac{(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}^j y_j}{1 - h_{j,j}}$.

Then, let's consider $\left(\mathbf{X}^{-jT} \mathbf{X}^{-j} \right)^{-1} \mathbf{X}^T \mathbf{y}$

$$\begin{aligned} \left(\mathbf{X}^{-jT} \mathbf{X}^{-j} \right)^{-1} \mathbf{X}^T \mathbf{y} &= \left(\left(\mathbf{X}^T \mathbf{X} \right)^{-1} + \frac{(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}^j \mathbf{x}^{jT} (\mathbf{X}^T \mathbf{X})^{-1}}{1 - \mathbf{x}^{jT} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}^j} \right) \mathbf{X}^T \mathbf{y} \\ &= \left(\mathbf{X}^T \mathbf{X} \right)^{-1} \mathbf{X}^T \mathbf{y} + \frac{(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}^j \mathbf{x}^{jT} (\mathbf{X}^T \mathbf{X})^{-1}}{1 - \mathbf{x}^{jT} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}^j} \mathbf{X}^T \mathbf{y} \\ &= \left(\mathbf{X}^T \mathbf{X} \right)^{-1} \mathbf{X}^T \mathbf{y} + \frac{(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}^j \mathbf{x}^{jT} (\mathbf{X}^T \mathbf{X})^{-1}}{1 - h_{j,j}} \mathbf{X}^T \mathbf{y} \\ &= \beta_{ls} + \frac{(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}^j \mathbf{x}^{jT} \beta_{ls}}{1 - h_{j,j}} \end{aligned}$$

$$\begin{aligned} \beta_{ls} - \beta_{ls}^{-j} &= \beta_{ls} - \left(\mathbf{X}^{-jT} \mathbf{X}^{-j} \right)^{-1} \mathbf{X}^T \mathbf{y} + \left(\mathbf{X}^{-jT} \mathbf{X}^{-j} \right)^{-1} \mathbf{x}^j y_j \\ &= \beta_{ls} - \left(\beta_{ls} + \frac{(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}^j \mathbf{x}^{jT} \beta_{ls}}{1 - h_{j,j}} \right) + \frac{(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}^j y_j}{1 - h_{j,j}} \\ &= - \frac{(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}^j}{1 - h_{j,j}} (\mathbf{x}^{jT} \beta_{ls} - y_j) \end{aligned}$$

Let's consider the nominator of D_j (Eq.(4))

$$\begin{aligned}
 \text{Nom}(D_j) &= (\beta_{ls} - \beta_{ls}^{-j})^T (\mathbf{X}^T \mathbf{X}) (\beta_{ls} - \beta_{ls}^{-j}) \\
 &= \left(\frac{(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}^j}{1 - h_{j,j}} (\mathbf{x}^{jT} \beta_{ls} - y_j) \right)^T (\mathbf{X}^T \mathbf{X}) \left(\frac{(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}^j}{1 - h_{j,j}} (\mathbf{x}^{jT} \beta_{ls} - y_j) \right) \\
 &= \frac{1}{(1 - h_{j,j})^2} (\mathbf{x}^{jT} \beta_{ls} - y_j)^T \mathbf{x}^{jT} (\mathbf{X}^T \mathbf{X})^{-1} (\mathbf{X}^T \mathbf{X}) (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}^j (\mathbf{x}^{jT} \beta_{ls} - y_j) \\
 &= \frac{1}{(1 - h_{j,j})^2} (\mathbf{x}^{jT} \beta_{ls} - y_j)^T \mathbf{x}^{jT} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}^j (\mathbf{x}^{jT} \beta_{ls} - y_j) \\
 &= \frac{h_{j,j}}{(1 - h_{j,j})^2} (\mathbf{x}^{jT} \beta_{ls} - y_j)^T (\mathbf{x}^{jT} \beta_{ls} - y_j) \\
 &= \frac{h_{j,j}}{(1 - h_{j,j})^2} (e_j^2)
 \end{aligned}$$

Recall that $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y} = \mathbf{y} - \mathbf{X}\beta_{ls}$.

- 1 Table of Contents
- 2 Outliers in training data
 - Outliers in measurements (y)
 - Leverage points in input features
- 3 Outlier diagnostics
 - Recap: linear regression
 - Hat matrix (unusual x values)
 - Residual analysis
 - Cook's distance
- 4 Robust regression
 - Weighted least-squares
 - Loss Functions for Robust Regression
 - Regression M-estimators
 - Median absolute deviation
 - Iterative re-weighted least square
- 5 Summary
- 6 Appendix
- 7 Index

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